Adoption of Organic Farming and Its Economic Viability

A Statistical Analysis using Farm-Level Data

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April 17, 2026

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Introduction

- This project studies the economic viability of organic farming compared to conventional farming using farm-level data
- The analysis focuses on key economic variables: yield, production cost, revenue, and net profit per hectare
- The objective is to evaluate whether lower input costs in organic farming compensate for potentially lower yields
- The study includes descriptive statistics, visualization, economic comparison, hypothesis testing, and regression analysis
- Crop-wise performance and the effect of long-term organic adoption on profitability are also examined
- The findings aim to provide insights for both farmers and policymakers regarding adoption decisions

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Dataset Description

- Total observations: 40 farms (20 Organic, 20 Conventional)
- Objective: Evaluate economic viability using net profit per hectare
- Key variables:
  - Production Cost (Rs./ha), Yield (ton/ha)
  - Market Price (Rs./ton)
  - Revenue (derived), Net Profit (main metric)
- Additional variable:
  - Years of Organic Farming (experience effect)

Dataset: Analytical Framework

- Profitability comparison: Organic vs Conventional (Net Profit/ha)
- Cost efficiency: Production cost and profit-to-cost ratio
- Crop-level analysis: Wheat, Rice, Vegetables
- Statistical methods:
  - Descriptive statistics and boxplots
  - Hypothesis testing (yield, profit)
  - Regression (profit vs yield, practice)
- Focus: Profitability and long-term adoption effects

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Problem 1: R Code

R Code

```
install.packages("csv")
library(csv)
data<-read.csv("Agridata.csv")
head(data)

data[,4:9] <- lapply(data[,4:9], function(x) {
  as.numeric(as.character(x))})

organic <- subset(data, data$Farming_Practice=="Organic")
conventional <- subset(data, data$Farming_Practice=="Conventional")

org_wheat <- subset(organic, organic$Crop_Type=="Wheat")
org_rice <- subset(organic, organic$Crop_Type=="Rice")
org_veg <- subset(organic, organic$Crop_Type=="Vegetables")

conv_wheat <- subset(conventional, conventional$Crop_Type=="Wheat")
conv_rice <- subset(conventional, conventional$Crop_Type=="Rice")
conv_veg <- subset(conventional, conventional$Crop_Type=="Vegetables")
```

Problem 1: R Code cont.

R Code

```
get_stats <- function(df) {
  data.frame(
    Mean_Yield = mean(df[,5]),
    Median_Yield = median(df[,5]),
    SD_Yield = sd(df[,5]),
    Mean_Cost = mean(df[,4]),
    Median_Cost = median(df[,4]),
    SD_Cost = sd(df[,4]),
    Mean_Revenue = mean(df[,7]),
    Median_Revenue = median(df[,7]),
    SD_Revenue = sd(df[,7]),
    Mean_Profit = mean(df[,8]),
    Median_Profit = median(df[,8]),
    SD_Profit = sd(df[,8])
  )
}
organic_stats <- get_stats(organic)
conventional_stats <- get_stats(conventional)
```

Problem 1: R Code cont..

R Code

```
org_rice_stats <- get_stats(org_rice)
conv_rice_stats <- get_stats(conv_rice)
org_wheat_stats <- get_stats(org_wheat)
conv_wheat_stats <- get_stats(conv_wheat)
org_veg_stats <- get_stats(org_veg)
conv_veg_stats <- get_stats(conv_veg)
final_table <- rbind(Organic = organic_stats,
                     Conventional = conventional_stats)

final_table_cropwise <- rbind(Org_Rice = org_rice_stats,
                              Conv_Rice = conv_rice_stats,
                              Org_Veg = org_veg_stats,
                              Conv_Veg = conv_veg_stats,
                              Org_Wheat = org_wheat_stats,
                              Conv_Wheat = conv_wheat_stats)
print(final_table)
print(final_table_cropwise)
cor(organic[,4:9])
cor(conventional[,4:8])
```

Problem 1: Results

Output

```
             Mean_Yield Median_Yield  SD_Yield Mean_Cost Median_Cost
Organic          3.1200        3.115 0.3967367   28346.5       28205
Conventional     3.7275        3.650 0.4589677   34300.0       34000

              SD_Cost Mean_Revenue Median_Revenue SD_Revenue Mean_Profit
Organic      4116.086     83403.20        82731.5   12011.28    55056.70
Conventional 3274.302     84843.75        85700.0   13331.60    50543.75

             Median_Profit SD_Profit
Organic              53724  11162.02
Conventional         49525  14388.59
```

Problem 1: Interpretation

- Summary statistics show distinct differences between organic and conventional farms
- Conventional farms achieve higher average yield (3.73 vs. 3.12 ton/ha) and exhibit slightly greater variability
- Organic farms operate with lower production costs (Rs 28,346 vs. Rs 34,300 per ha), though cost variability is higher among organic farms
- Revenue levels are similar across both systems, with conventional farms having a marginally higher mean revenue
- Organic farms outperform in profitability, with higher mean net profit (Rs 55,057 vs. Rs 50,544 per ha) and higher median profit
- Profit variability is greater for conventional farms
- Organic farming involves lower yield but also lower cost, resulting in higher average profitability

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Problem 2: R Code

R Code

```
library(ggplot2)
library(dplyr)
library(tidyr)

df <- read.csv("Agridata.csv")
df$Farming_Practice <- as.factor(df$Farming_Practice)
ggplot(df, aes(x = Farming_Practice,
               y = Yield_ton_per_ha,
               fill = Farming_Practice)) +
geom_boxplot(alpha = 0.7) +
labs(
title = "Comparison of Yield by Farming Type",
x = "Farming Practice",
y = "Yield (tons per hectare)"
) +
theme_minimal() +
theme(legend.position = "none")
```

Problem 2: R Code Cont.

R Code

```
ggplot(df, aes(x = Farming_Practice,
y = Production_Cost_per_ha,
fill = Farming_Practice)) +
geom_boxplot(alpha = 0.7) +
labs(
title = "Comparison of Production Cost by Farming Type",
x = "Farming Practice",
y = "Cost (per hectare)"
) +
theme_minimal() +
theme(legend.position = "none")
```

Problem 2: R Code Cont..

R Code

```
ggplot(df, aes(x = Farming_Practice,
y = Net_Profit_per_ha,
fill = Farming_Practice)) +
geom_boxplot(alpha = 0.7) +
labs(
title = "Comparison of Net Profit by Farming Type",
x = "Farming Practice",
y = "Net Profit (per hectare)"
) +
theme_minimal() +
theme(legend.position = "none")
```

Problem 2: Yield Comparison

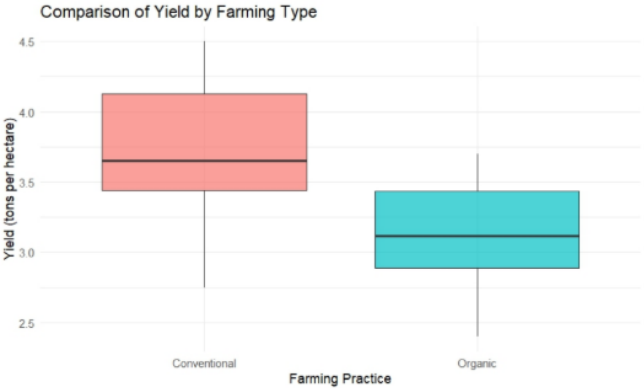


Figure 1: Boxplot comparing yield across farming practices

Figure 1

Problem 2: Production Cost Comparison

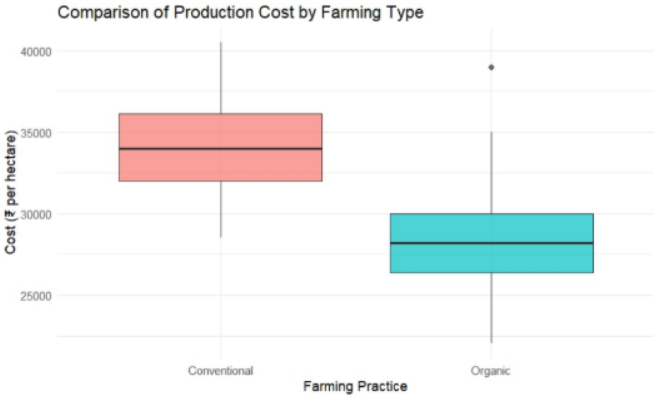


Figure 2: Boxplot comparing production cost across farming practices

Figure 2

Problem 2: Net Profit Comparison

Figure 3: Boxplot comparing net profit across farming practices

Problem 2: Interpretation

- Boxplots show clear differences between farming practices
- Conventional farms have a higher median yield and greater spread, indicating higher but more variable output
- Organic farms exhibit lower, more consistent yields
- Production costs are higher (although less dispersed) for conventional farms
- Organic farms maintain lower costs
- For net profit, organic farms have a higher median with a lesser spread
- This indicates better and less variable profitability

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Problem 3: R Code

R Code

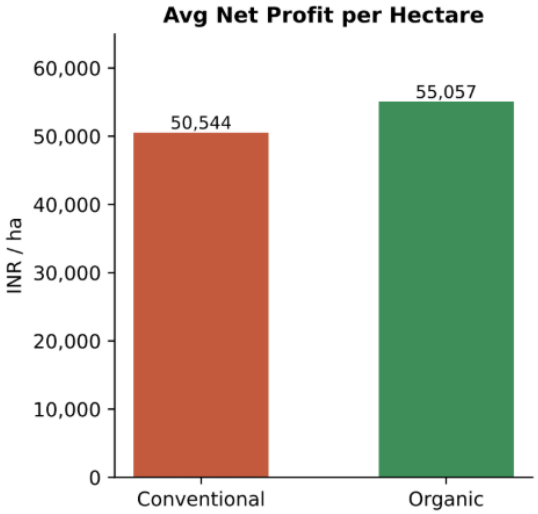
```
suppressPackageStartupMessages({
  library(dplyr)
  library(ggplot2)
  library(gridExtra)
  library(grid)
  library(scales)})
df <- read.csv("Agridata.csv", stringsAsFactors = FALSE)
overall <- df %>%
  group_by(Farming_Practice) %>%
  summarise(
    Farms = n(),
    Avg_Cost_per_ha = mean(Production_Cost_per_ha),
    Avg_Yield = mean(Yield_ton_per_ha),
    Avg_Price_per_ton = mean(Market_Price_per_ton),
    Avg_Revenue_per_ha = mean(Total_Revenue_per_ha),
    Avg_Profit_per_ha = mean(Net_Profit_per_ha),
    .groups = "drop"
  ) %>%  mutate(Profit_to_Cost_Ratio =
         Avg_Profit_per_ha / Avg_Cost_per_ha)
```

Problem 3: R Code Cont.

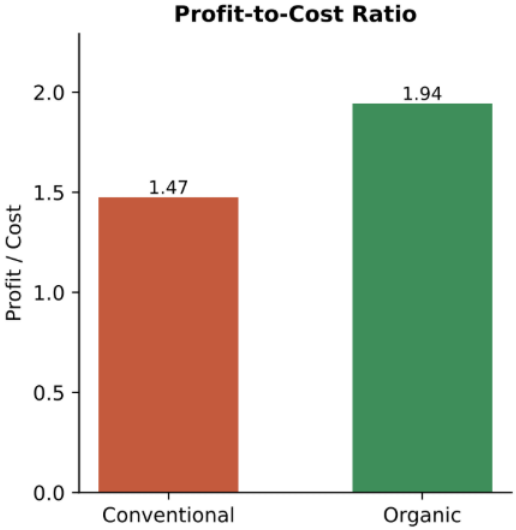
R Code

```
by_crop <- df %>%
  group_by(Crop_Type, Farming_Practice) %>%
  summarise(
    Farms = n(),
    Avg_Cost_per_ha = mean(Production_Cost_per_ha),
    Avg_Profit_per_ha = mean(Net_Profit_per_ha),
    .groups = "drop"
  ) %>%
  mutate(Profit_to_Cost_Ratio =
         Avg_Profit_per_ha / Avg_Cost_per_ha)
winners <- by_crop %>%
  group_by(Crop_Type) %>%
  summarise(
    Winner_by_Profit =
      Farming_Practice[which.max(Avg_Profit_per_ha)],
    Winner_by_PCR =
      Farming_Practice[which.max(Profit_to_Cost_Ratio)],
    .groups = "drop" )
```

Problem 3: Avg Net Profit



Problem 3: Profit-to-Cost Ratio



Problem 3: Crop-wise Profit

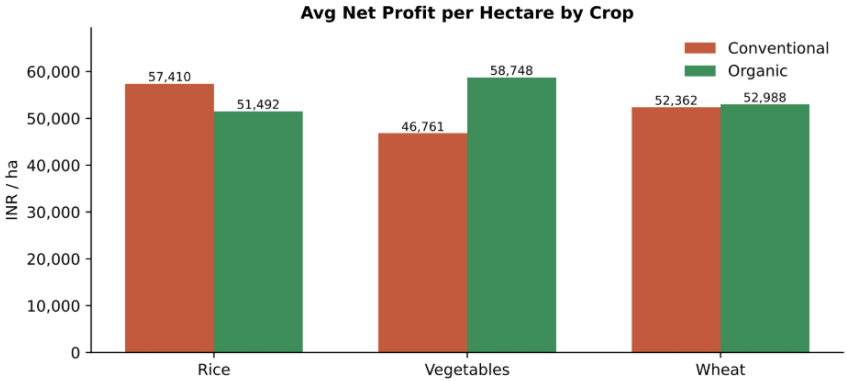


Figure 6: Avg Net Profit per Hectare by Crop

Problem 3: Crop-wise Ratio

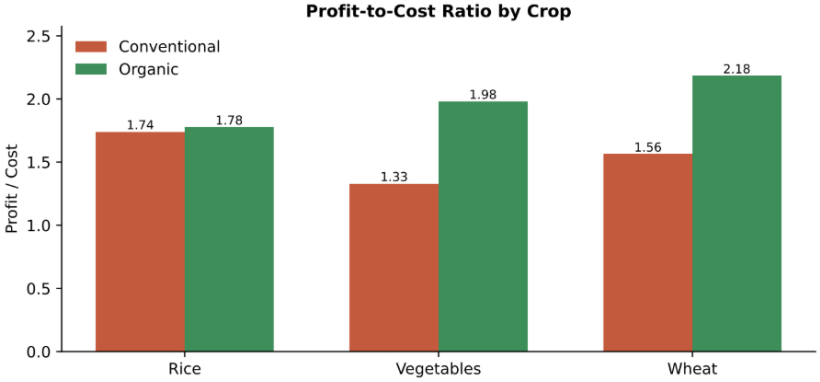


Figure 7: Profit-to-Cost Ratio by Crop

Problem 3: Interpretation

- Organic farming outperforms conventional farming on both profitability (mean profit) and efficiency (profit-to-cost ratio)
- Average net profit is higher for organic farms (Rs 55,057 vs. Rs 50,544 per ha), driven by lower production costs (Rs 28,346 vs. Rs 34,300 per ha)
- The profit-to-cost ratio further highlights this advantage (1.94 for organic vs. 1.47 for conventional)
- Crop-wise, organic farming shows clear advantages in most cases
- Vegetables: Higher profit (Rs 58,748 vs. Rs 46,761) and stronger efficiency (1.98 vs. 1.33)
- Wheat: Similar profit levels (Rs 52,988 vs. Rs 52,362), but significantly higher efficiency (2.18 vs. 1.56) due to lower costs
- Rice: Conventional has higher profit (Rs 57,410 vs. Rs 51,492), though efficiency is nearly identical (1.74 vs. 1.78)
- Overall, organic farming delivers higher returns per unit cost and is economically more viable, particularly for vegetables and wheat

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Problem 4: R Code

R Code

```
df <- read.csv("Agridata.csv")
df$Farming_Practice <- as.factor(df$Farming_Practice)
t_profit <- t.test(Net_Profit_per_ha ~ Farming_Practice, data = df)
t_yield <- t.test(Yield_ton_per_ha ~ Farming_Practice, data = df)
cat("Net Profit p-value:", t_profit$p.value, "\n")
cat("Yield p-value:", t_yield$p.value, "\n")
cat("Net Profit:", ifelse(t_profit$p.value < 0.05,
"Significant", "Not Significant"), "\n")
cat("Yield:", ifelse(t_yield$p.value < 0.05,
"Significant", "Not Significant"), "\n")
w_profit <- wilcox.test(Net_Profit_per_ha ~ Farming_Practice, data = df)
w_yield <- wilcox.test(Yield_ton_per_ha ~ Farming_Practice, data = df)
w_profit$p.value
w_yield$p.value
```

Problem 4: Results

- Net Profit (t-test): $p = 0.2751$ (Not Significant)
- Net Profit (Wilcoxon): $p = 0.2314987$ (Not Significant)
- Yield (t-test): $p = 0.000069$ (Significant)
- Yield (Wilcoxon): $p = 0.0003716679$ (Significant)

Problem 4: Interpretation

- Two independent sample tests were conducted at $\alpha = 0.05$ to evaluate differences in yield and net profit
- For net profit, t-test p-value = 0.2751 (not statistically significant)
- Wilcoxon test p-value = 0.2314987 confirms the same result
- No significant difference in net profit between organic and conventional farming
- This aligns with the overlap observed in boxplot distributions
- For yield, t-test p-value = 0.000069 (statistically significant)
- Wilcoxon test p-value = 0.0003716679 supports this finding
- Farming practice significantly affects yield, but not net profit

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Problem 5: R Code

R Code

```
farmdata_df <- read.csv("Agridata.csv")
organic_data <- farmdata_df[
farmdata_df$Farming_Practice == "Organic",]
corr_val_pearson <- cor(
organic_data$Years_of_Organic_Farming,
organic_data$Net_Profit_per_ha,
method = "pearson")
corr_val_pearson
farmdata_df$Farming_Practice <- factor(
farmdata_df$Farming_Practice,
levels = c("Conventional","Organic"))
model_1 <- lm(Net_Profit_per_ha ~
Yield_ton_per_ha + Farming_Practice,
data = farmdata_df)
summary(model_1)
```

Problem 5: Results

Output

```
corr_val_pearson
[1] 0.1723398

Coefficients:
(Intercept)              -41606
Yield_ton_per_ha          24722
Farming_PracticeOrganic   19531

R-squared: 0.6884
```

Problem 5: Interpretation

- The correlation between years of organic farming and net profit is weakly positive ($r = 0.172$), indicating that longer experience in organic farming is associated with only a marginal increase in profit
- The low magnitude suggests no meaningful linear relationship
- The regression model explains a substantial share of variation in profit ($R^2 = 0.688$)
- Yield is the dominant predictor: a one-unit increase in yield raises profit by Rs 24,722, holding farming practice constant
- Farming practice is also significant—organic farms earn Rs 19,531 more per hectare than conventional farms at the same yield
- Both predictors are statistically significant ($p < 0.001$), confirming that yield drives profitability, while organic practice provides an additional positive effect

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Economic Viability of Organic Farming

- Organic farming is economically viable and, on average, more efficient than conventional farming
- Although yields are lower, substantially reduced input costs result in higher average profits
- Organic farming shows a superior profit-to-cost ratio
- Hypothesis testing shows that profit differences are not statistically significant
- This indicates that the advantage is not consistent across all farms

Crop-wise Performance under Organic Farming

- Organic farming performs strongest in vegetables
- Profit: Rs 58,748 vs Rs 46,761 and efficiency: 1.98 vs 1.33
- Wheat favors organic farming through comparable profits but significantly better cost efficiency
- In rice, conventional farming achieves slightly higher profit
- Efficiency in rice remains similar across both practices

Effect of Long-term Organic Adoption

- Long-term adoption does not significantly improve profitability
- Correlation between years of organic farming and profit is weakly positive
- Experience in organic farming does not translate into strong economic gains

Recommendations for Farmers

- Adoption of organic farming for vegetables and wheat where returns and efficiency are consistently higher
- For rice, adoption should be done selectively based on market price premiums and risk tolerance
- Focus on yield improvement within organic systems, as profit is highly yield-driven

Recommendations for Policymakers

- Promote organic farming in high-return crops (vegetables, wheat) rather than uniform adoption
- Improve market access and price realization for organic produce to stabilize returns
- Support extension services targeting yield improvement in organic systems

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Conclusion

- Organic farming is economically viable and generally more efficient than conventional farming
- Lower production costs play a key role in offsetting lower yields
- Profit differences between farming practices are not statistically significant
- Yield is significantly affected by farming practice and remains the primary driver of profitability
- Organic farming performs better for vegetables and wheat, while rice shows mixed results
- Long-term organic adoption does not significantly improve profitability
- Overall, organic farming provides higher returns per unit cost but with some variability across farms

Contents

1 Introduction

2 Dataset Description

3 Problem 1: Data Exploration and Descriptive Statistics

4 Problem 2: Data Visualization

5 Problem 3: Economic Viability Analysis

6 Problem 4: Hypothesis Testing

7 Problem 5: Correlation and Regression

8 Final Comments

9 Conclusion

10 Thank You

Thank You

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